

 Marwadi University Marwadi Chandarana Group 	Marwadi University Faculty of Engineering & Technology Department of Information and Communication Technology	
Subject: Digital Signal and Image Processing (01CT1513)	AIM: Simulate auto-correlation and cross-correlation on discrete time signals	
Experiment No: 3	Date:30-07-2026	Enrollment No:92400133121

AIM: Simulate cross correlation and autocorrelation on discrete time signals.

Theory:

- Cross-correlation and autocorrelation are mathematical operations used to measure the similarity or correlation between two signals.
They are widely used in various applications, such as signal processing, image processing, and pattern recognition.
- Cross-correlation measures the similarity between two signals at different time shifts.
It computes the dot product of one signal with a time-shifted version of the other signal.
The resulting cross-correlation signal indicates the similarity between the two signals at different time lags.
- Autocorrelation, on the other hand, measures the similarity of a signal with a time-shifted version of itself.
It computes the cross-correlation of a signal with itself.
The autocorrelation signal shows how the signal is correlated with itself at different time lags.

Flowchart:

- ✓ The flowchart for the program will consist of the following steps:
- ✓ Define two discrete-time signals (signal 1 and signal 2) or use the same signal for autocorrelation.
- ✓ Compute the cross-correlation of the two signals or the autocorrelation of a single signal.
- ✓ Plot and display the cross-correlation or autocorrelation signal.
- ✓ Save the cross-correlation or autocorrelation signal (optional).

Code:

```
import numpy as np
import matplotlib.pyplot as plt

def cross_correlation(signal1, signal2):
    # cross-correlation
    cross_corr = np.correlate(signal1, signal2, mode='full')
    return cross_corr

def auto_correlation(signal):
    # auto-correlation
    auto_corr = np.correlate(signal, signal, mode='full')
    return auto_corr

signal1 = np.array([1,2,3,2])
signal2 = np.array([2,1,1,2])
```

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```
# cross-correlation
cross_corr = cross_correlation(signal1, signal2)
# auto-correlation
auto_corr = auto_correlation(signal1)
```

```
plt.figure(figsize=(10, 6))
```

```
plt.subplot(2, 1, 1)
plt.stem(cross_corr)
plt.title('Cross Correlation')
plt.xlabel('Time lag')
plt.ylabel('Magnitude')
```

```
plt.subplot(2, 1, 2)
plt.stem(auto_corr)
plt.title('Auto Correlation')
plt.xlabel('Time lag')
plt.ylabel('Magnitude')
```

```
plt.tight_layout()
plt.show()
```

Output:

